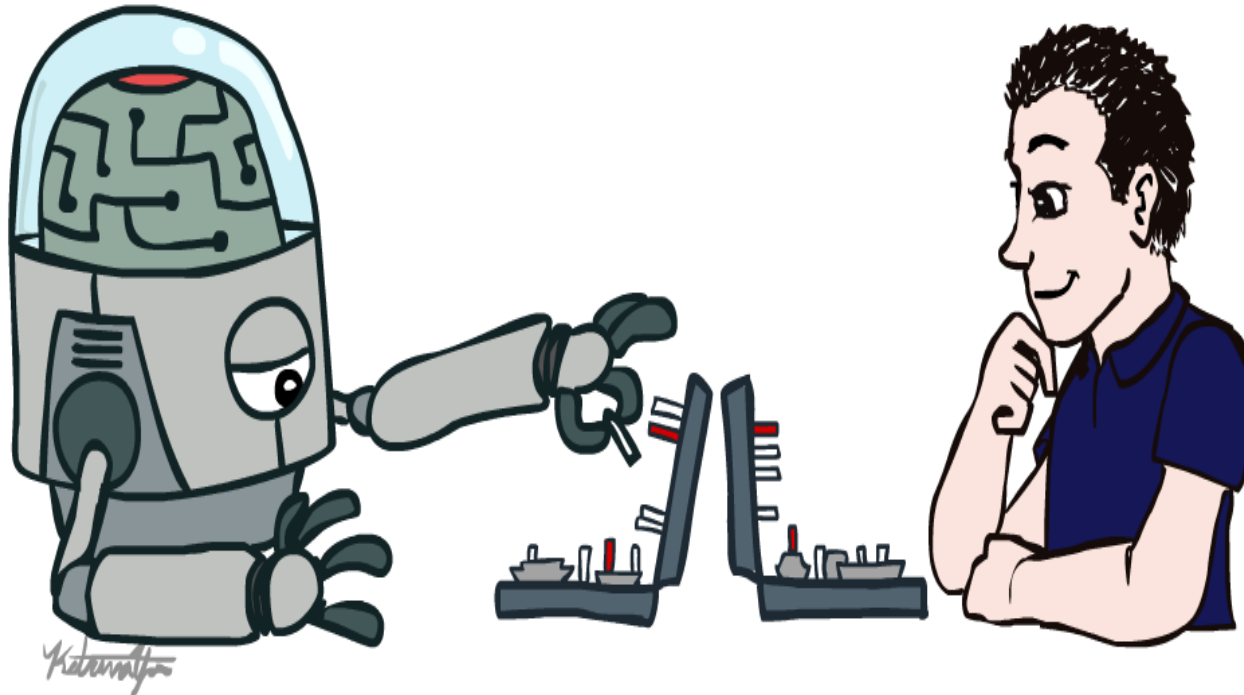


# Introduction to Artificial Intelligence



# **Chapter 1**

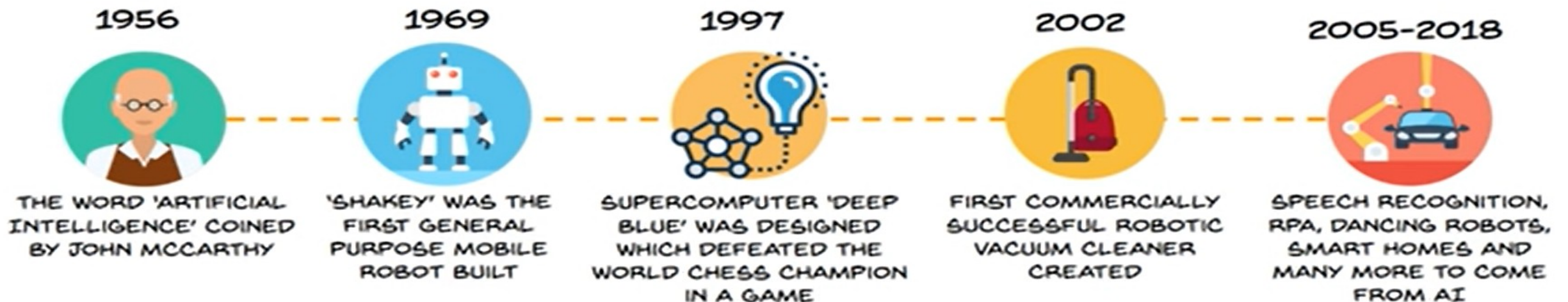
## **Introduction to Artificial Intelligence**

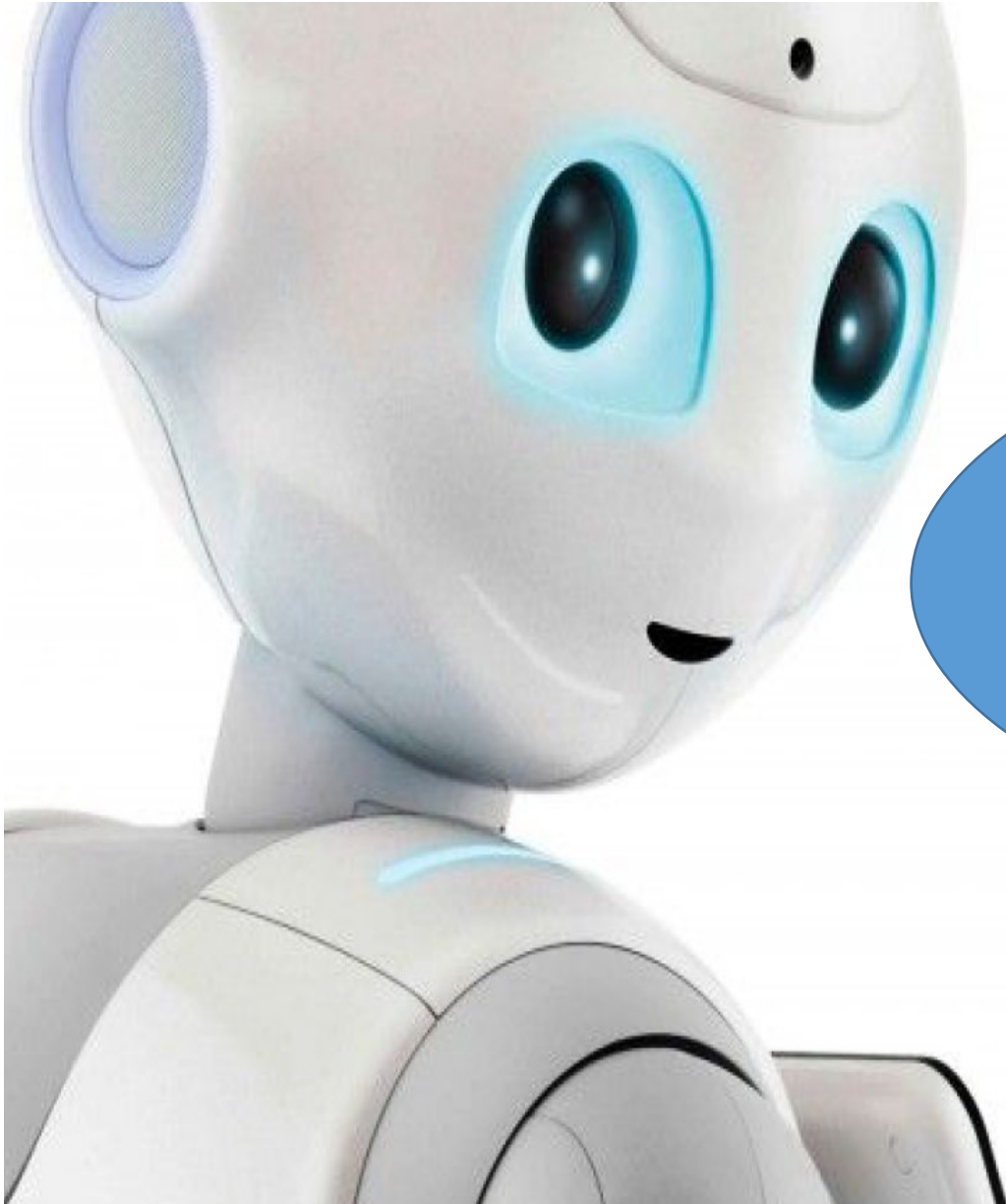
# Main Contents

- History of AI, an overview
- Definitions of AI
- Goals of AI
- AI approaches
- Branches of AI
- Application of AI

# History of AI, an overview

- What did you learn from evolutionary history of AI? , see the **compression** of time in the diagram.
- In 1956 AI was just an idea. **John McCarthy**, who coined the term in 1956, defines it as "*the science and engineering of making intelligent machines, especially intelligent computer programs.*"
- AI is an old approach, but now different AI based technologies are coming out almost daily/monthly in the world of AI. ***So what is AI? What is the demand of AI?***





**What is Artificial  
Intelligence (AI) ?**

# Definitions of AI

- **Artificial Intelligence** is the branch of computer science concerned with making computers behave like humans.
- Most AI textbooks define artificial intelligence as:
  - "*the study and design of intelligent agents*," where an **intelligent agent** is a system that **perceives** its **environment** and **takes actions** which maximize its chances of success.



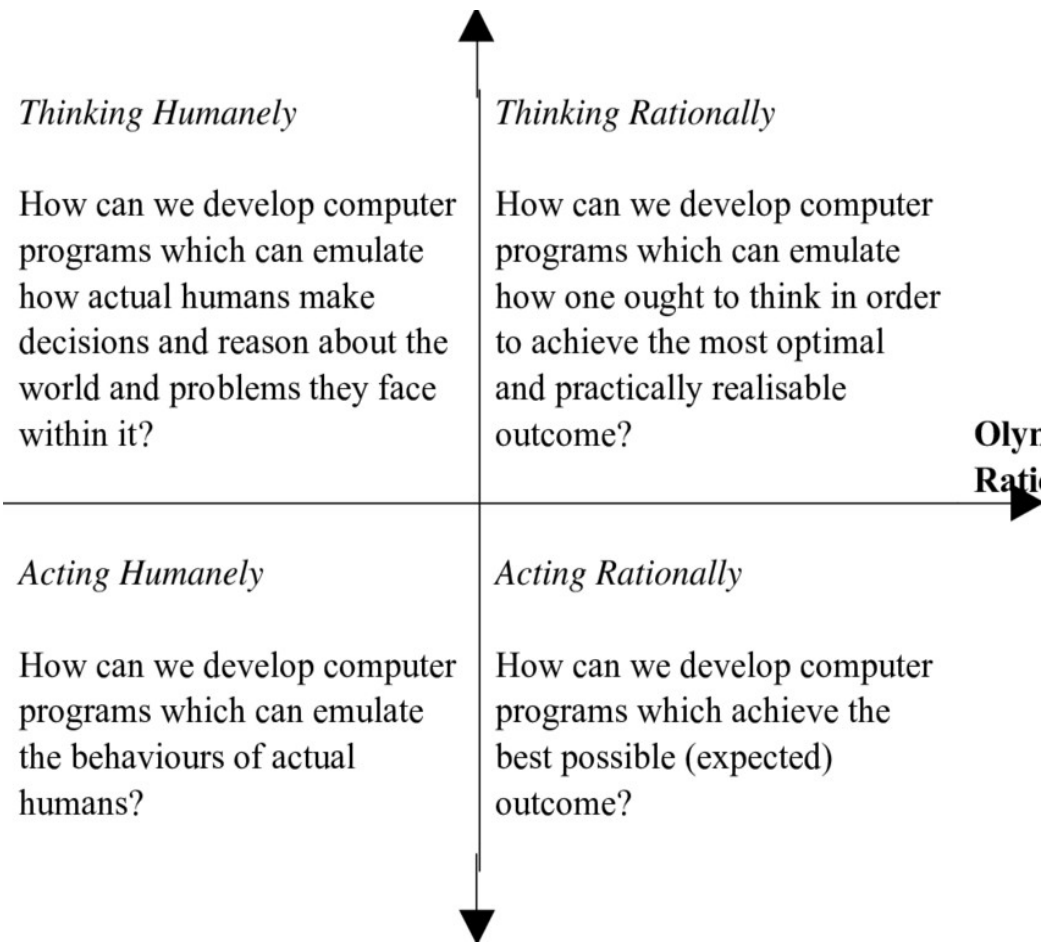
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- The definitions of AI according to some text books are categorized into four approaches and are summarized in the table below :

Table 1: Four Categories of AI Definitions

|                              | Human Behaviour                                                                       | Rational Behaviour                                       |
|------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------|
| Thinking<br>(Mental Process) | 1. Thinking Humanly<br>Machines that think intelligently like humans                  | 3. Thinking Rationally<br>Machines that think rationally |
| Acting<br>(Action)           | 2. Acting Humanly<br>Machines that perform activities that human consider intelligent | 4. Acting Rationally<br>Machines that act rationally     |

Source: Adapted from Russell and Norvig (2010), Figure 1.1, p.2.



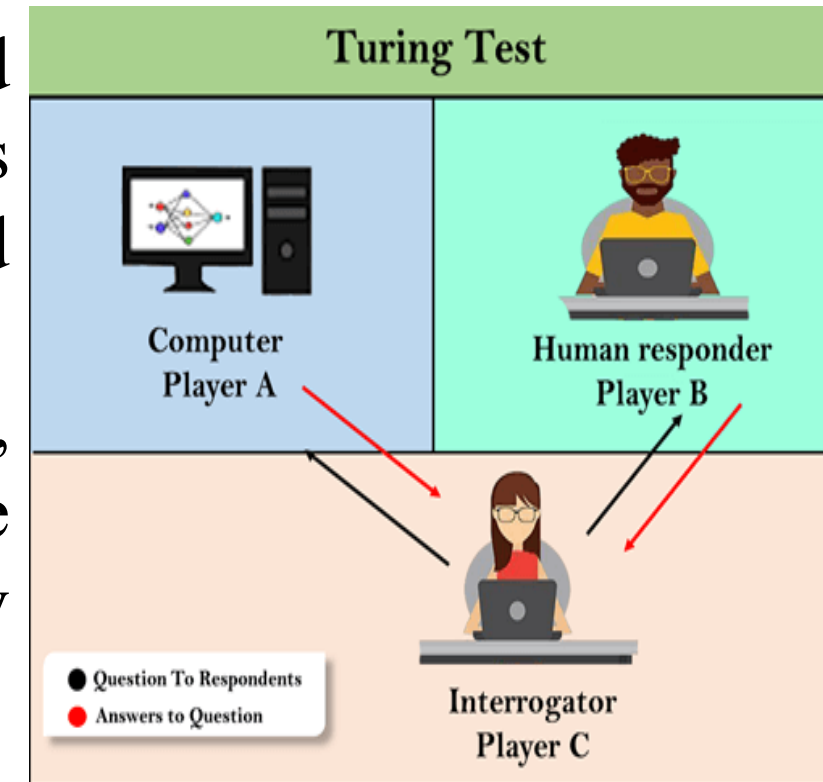
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## Classic example -

- In 1950, Alan Turing introduced a test to check whether a machine can **think like a human** or not, this test is known as the **Turing Test**.
- This game involves three players in which one player is Computer, another player is human responder, and the third player is a human Interrogator, who is isolated from other two players and his job is to find that which player is machine among two of them.
- Consider, *Player A* is a computer, *Player B* is human, and *Player C* is an interrogator. Interrogator is aware that one of them is machine, but he needs to identify this on the basis of questions and their responses.

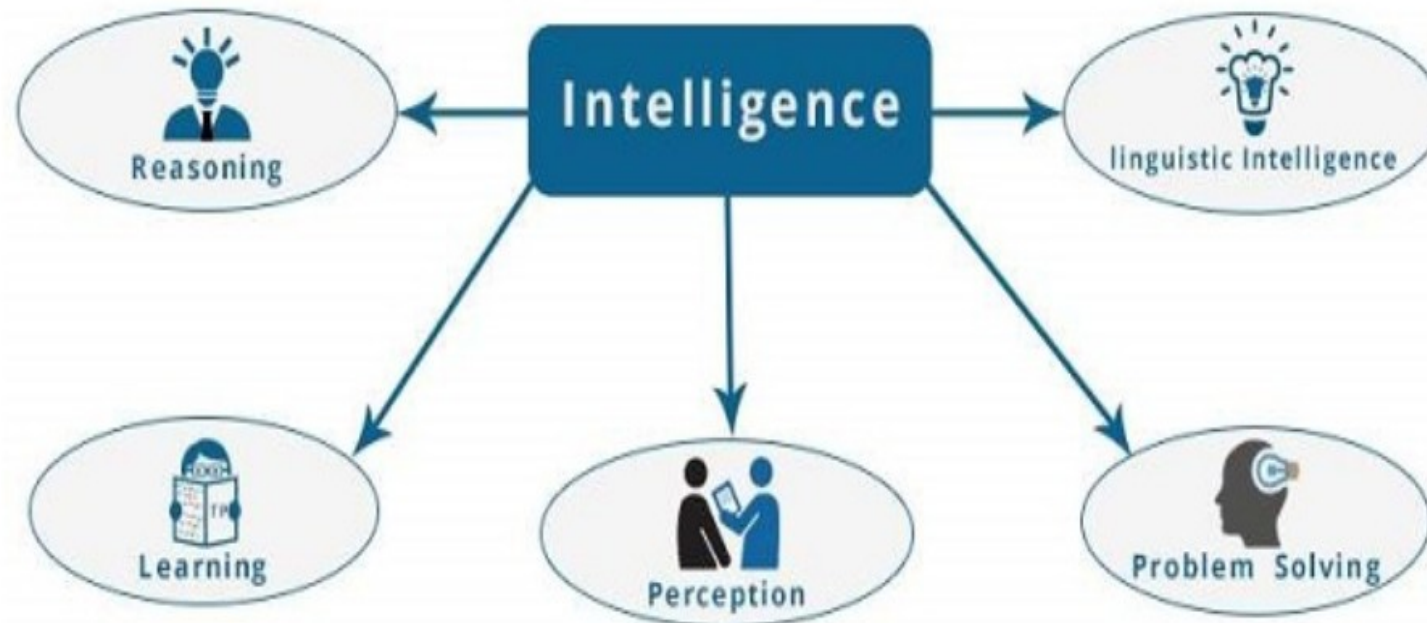


**Alan Turing**



# Compositions of intelligence

- The intelligence is intangible. It is composed of –
  - **Reasoning** : draw a conclusion from the situation given to it. Eg . Imaging chees playing with a robot
  - **Learning** :It is the activity of gaining knowledge or skill by studying, practicing, or experiencing something. Eg. Shake hand sign.

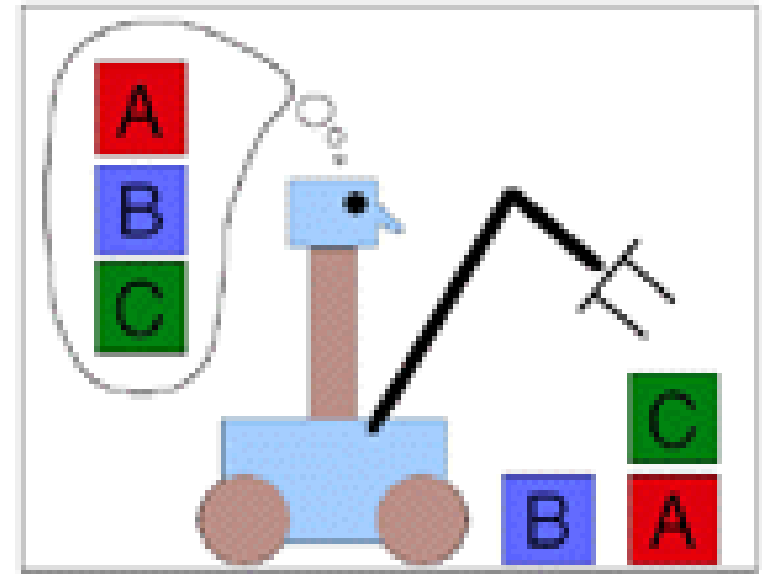


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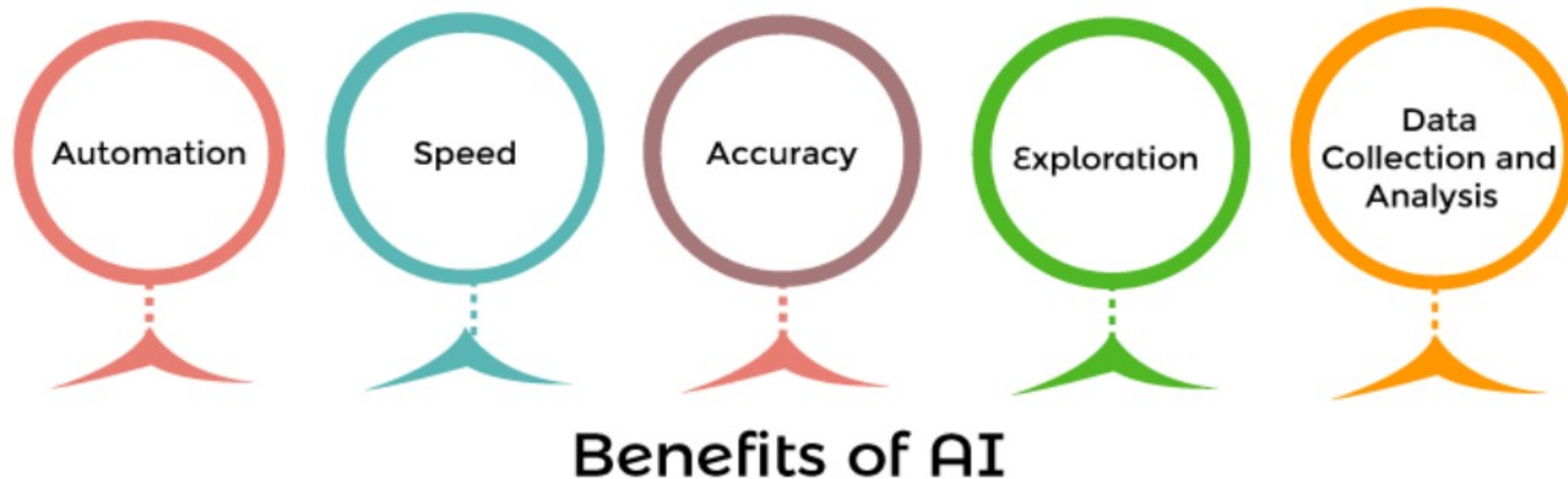
- **Problem Solving:** to search to all possible actions to reach the goal given. Eg . Imaging chees playing with a robot
- **Perception:** It is the process of acquiring, interpreting, selecting, and organizing sensory information.
  - For AI has to scan and understand its environment. Eg object recognition in AI system requires objects from all direction.
- **Linguistic Intelligence :** one's ability to use, comprehend, speak, and write the verbal and written language.
  - Important for interpersonal communication.

# Goals of AI

- To make computers more useful by letting them take over dangerous or tedious tasks from human
- Understand principles of human intelligence and **Implement Human Intelligence in Machines:** Creating systems that understand, think, learn, and behave like humans

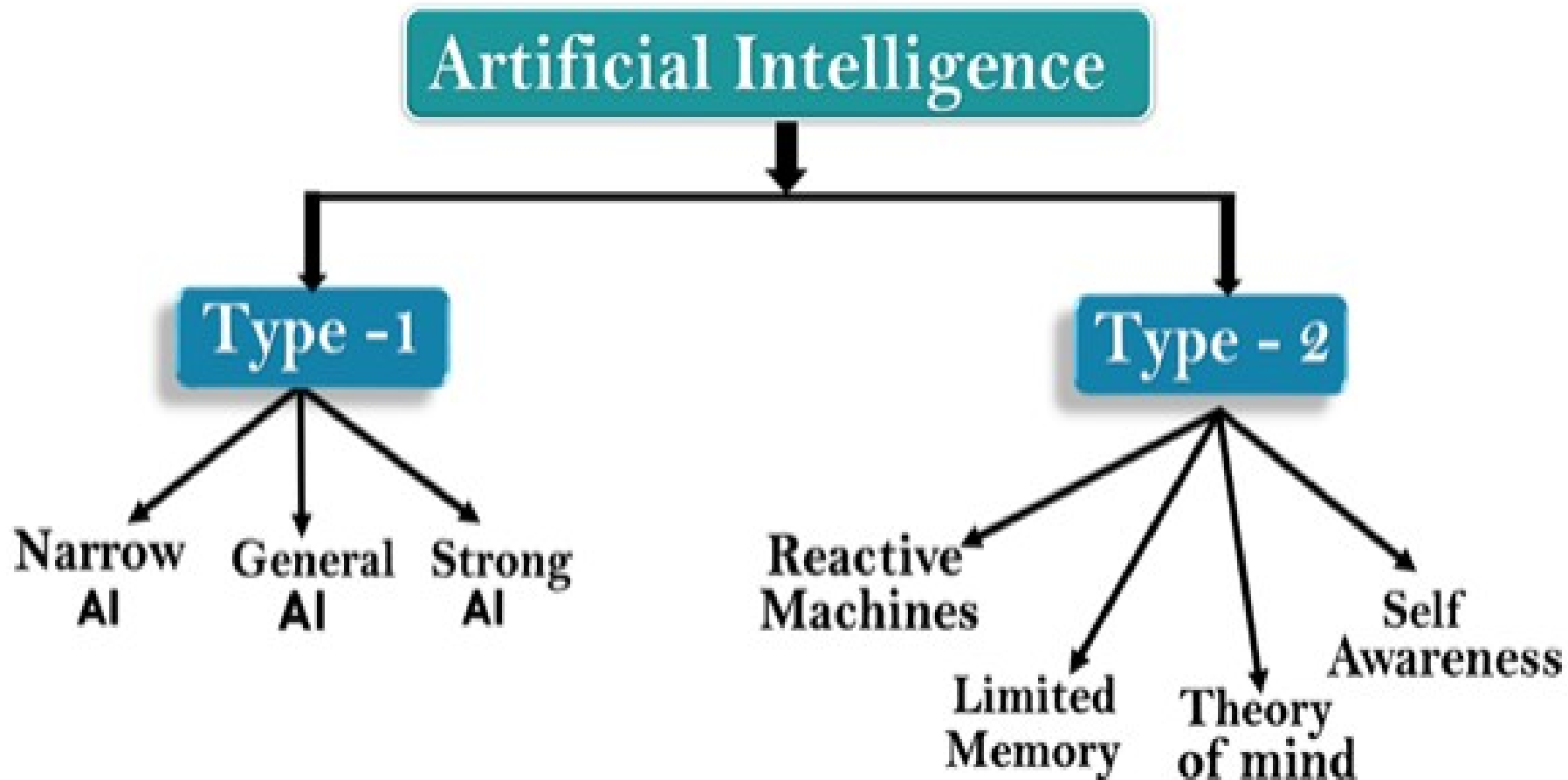


- AI has various uses in the modern-day scenario.
- Industries are using AI to automate processes, and better AI algorithms are being developed every day to speed up various industry processes/tasks.
- The most important purpose of AI is to reduce human casualties in  
➔ Wars, Dangerous Workspaces, Car Accidents, etc.

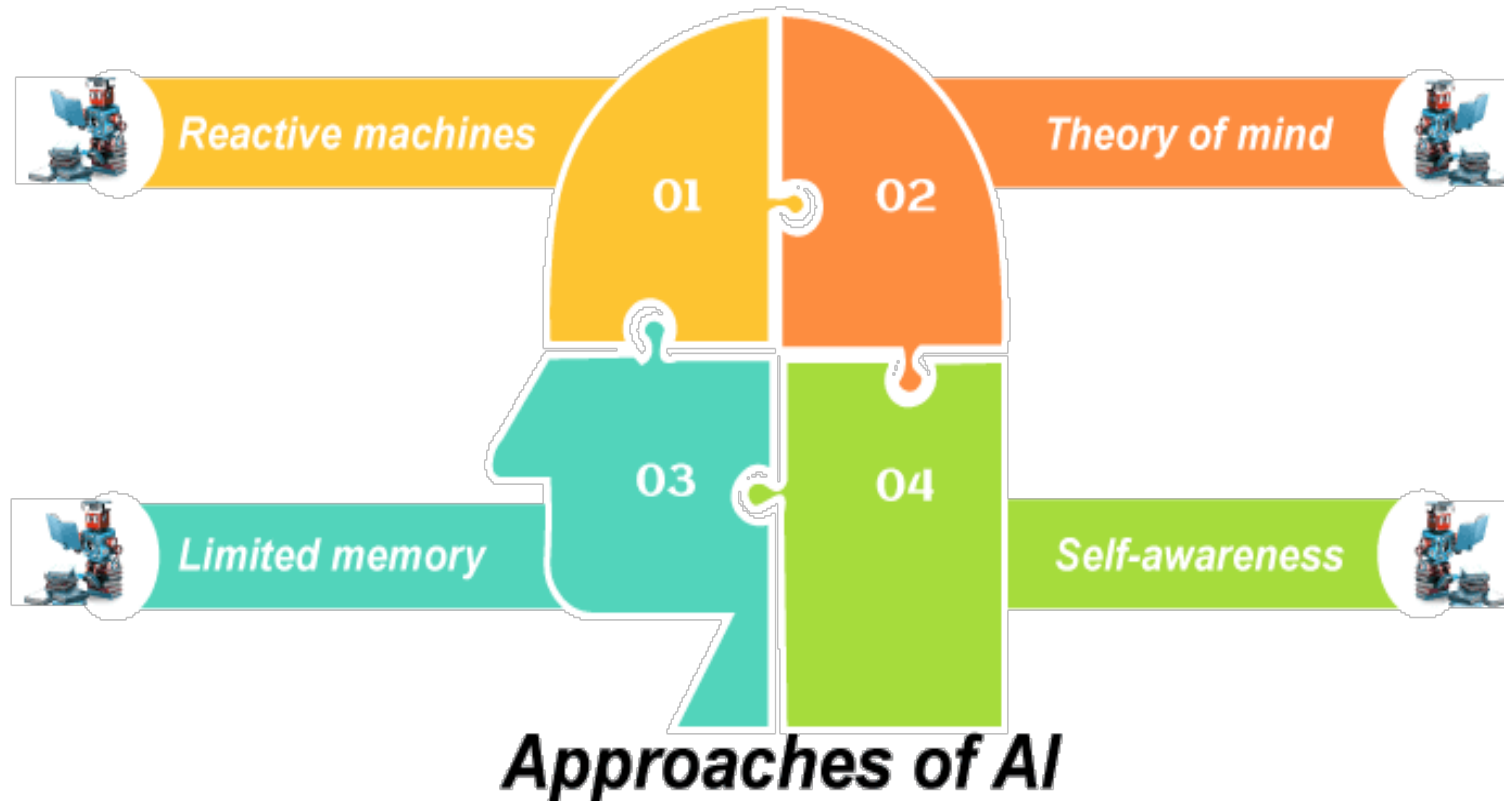


# AI Approaches

- AI can be broadly categorized into 2 groups, based on their **ability** and **functionality**, however both of these have their own sub-types.



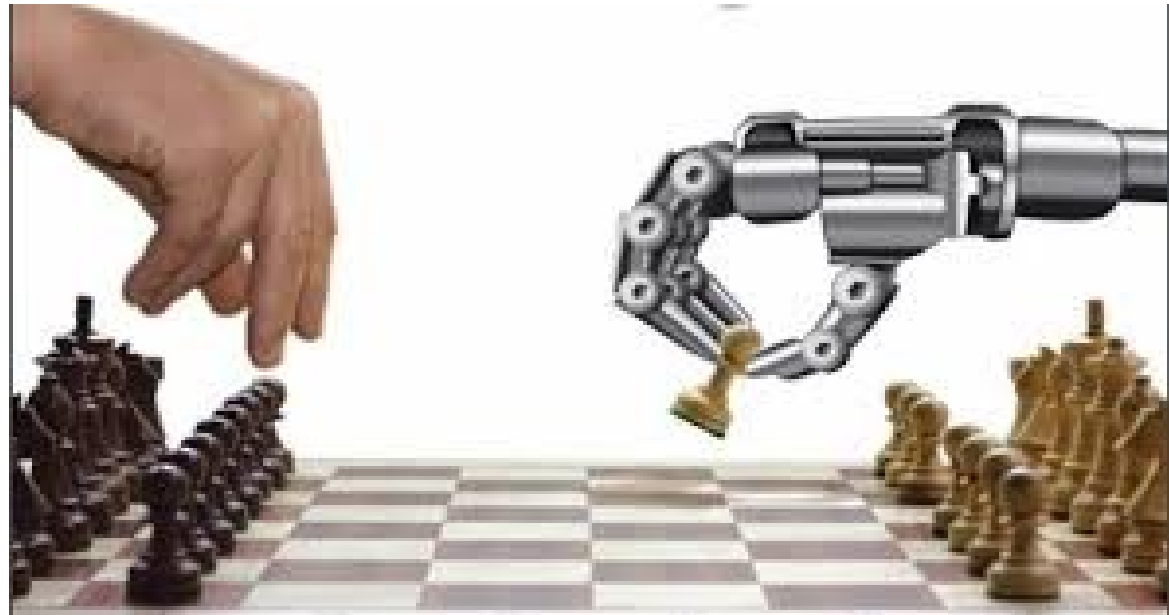
- Four types of artificial intelligence approaches based on how machines behave **(functionality)** - reactive machines, limited memory, theory of mind, and self-awareness.



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## 1. Reactive machines

- These machines are the most basic form of [AI applications](#).
  - E.g. **Deep Blue**, IBM's chess-playing supercomputer, etc.
- [AI](#) teams *do not use training sets* to feed the machines or store subsequent data for future references. Based on the move made by the opponent, the machine decides/predicts the next move.



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## 2. Limited memory

- These machines belong to the Category II category of **AI applications**, and Self-propelled cars are the perfect example.
- Over time, **these machines are fed with data** and trained on the speed and direction of other **cars, lane markings, traffic lights, curves of roads**, and other important factors.



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## 3. Theory of mind

- As the name sounds, the theory of mind machines represents an advanced class of technology and exists only as a concept.
- It is a concept where bots will understand and react to human emotions, thoughts.

## 4. Self-awareness

- Self-aware AI is the exact thing that is portrayed in AI-movies.
- The critical AI robots that think on their own and destroy humans are the ideology-driven from self-aware AI.
- It is the stage where AI teams build machines with self-awareness factors programmed into them.

# .....Cont'd

- Four types of artificial intelligence approaches **based on ability** :

## 1. Artificial Narrow Intelligence (ANI)

- Also known as '**weak AI**' is one of the most frequently experienced types of AI.
- Every AI machine we use and see today is from this field, which operates under a limited set of constraints.
- E.g., voice recognition AI is used to predict people's voice based on the dataset it is trained with

## 2. Artificial General Intelligence (AGI)

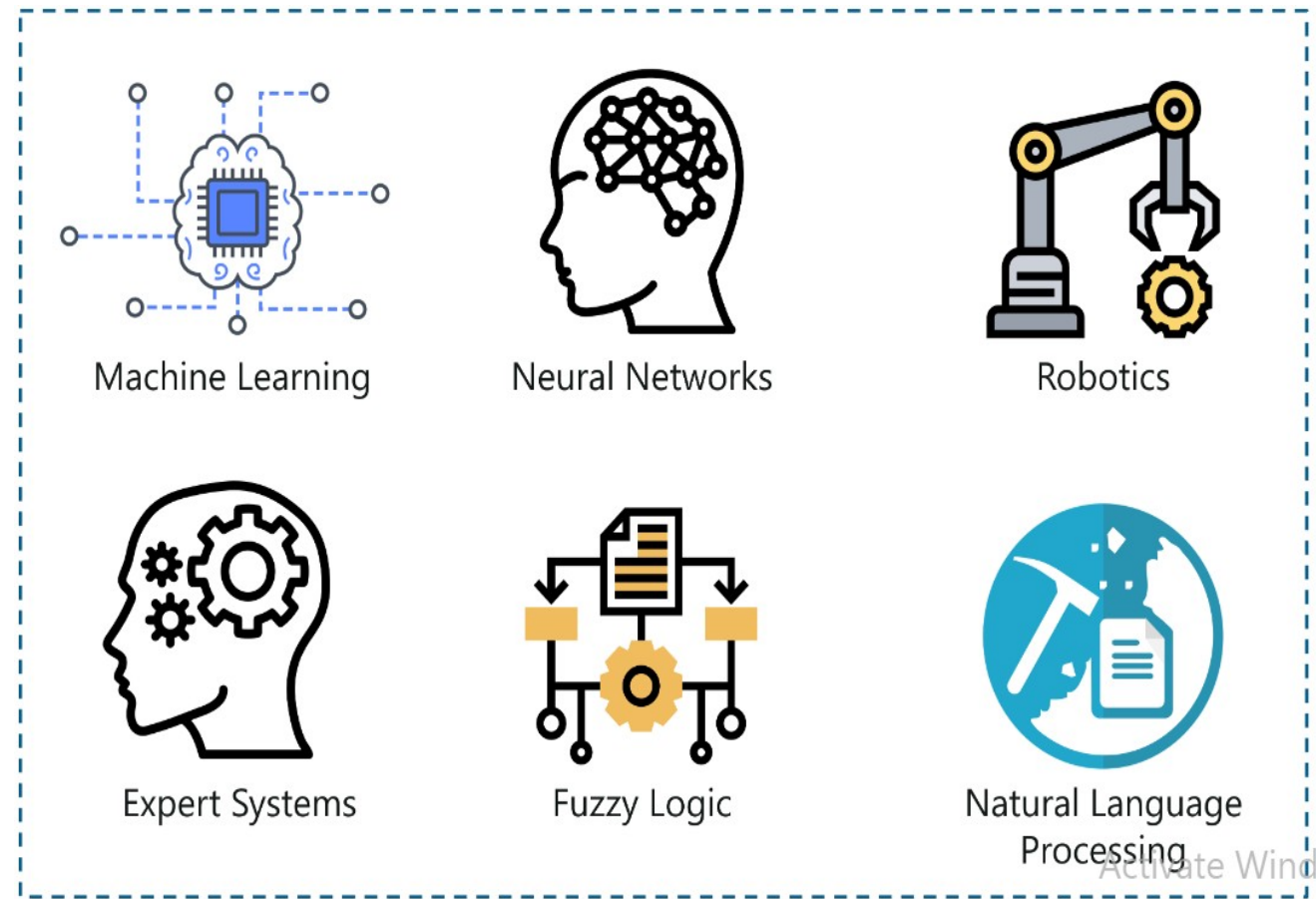
- Also known as '**strong AI**' allows a machine to apply knowledge and skills in different contexts.
- This more closely mirrors human intelligence by providing opportunities for autonomous learning and problem-solving. In other words, AGI can successfully perform any intellectual task that a human being can

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### 3. Artificial Super Intelligence (ASI)

- Artificial Super Intelligence is something more advanced than ‘strong AI’ which could end the human race.
- They will not only replicate the multi-faceted intelligence of human beings but will be exceedingly better at everything.
- The ASI will be designed with better memory, faster data processing and analysis, and decision-making capabilities.

- Artificial Intelligence can be used to solve real-world problems by implementing the following processes/ techniques:
  - Machine Learning
  - Deep Learning
  - Natural Language Processing
  - Robotics
  - Expert Systems
  - Fuzzy Logic



### 1. Machine Learning

### 2. Deep Learning

### 3. Natural Language Processing

- Natural Language Processing (NLP) refers to the science of drawing insights from natural human language in order to communicate with machines and grow businesses.
  - Twitter uses NLP to filter out terroristic language in their tweets, Amazon uses NLP to understand customer reviews and improve user experience.

### 4. Robotics

- They are artificial agents acting in a real-world environment to produce results by taking accountable actions.
  - *Sophia* the humanoid is a good example of AI in robotics.

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## **5. Fuzzy Logic**

- A technique that represents and modifies uncertain information by measuring the degree to which the hypothesis is correct. Fuzzy logic is also used for reasoning about naturally uncertain concepts.
- Used in the medical fields to solve complex problems that involve decision making.

## **6. Expert Systems**

- An expert system is an AI-based computer system that learns and reciprocates the decision-making ability of a human expert.
- Mainly used in information management, medical facilities, loan analysis, virus detection and so on.

# Applications of Artificial intelligence



**End of Chapter**